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6.4 OLAP Cube Creation with Facts and Dimensions

What we are going *to do?*

Understand Data Structure for Multi-Dimensional Analysis.

Gain foundational knowledge in OLAP cube creation.

Understand the purpose and structure of OLAP cubes in data analysis.

Define and differentiate facts and dimensions in OLAP.

Learn how to set up a basic OLAP cube for multi-dimensional analysis.

6.4.1 Online Analytical Processing (OLAP)

What is OLAP?

Online Analytical Processing

- **A technology for organizing and analyzing data from multiple perspectives.**
- **Supports complex queries and enables quick data retrieval.**
- **Essential for Business Intelligence (BI) applications and Decision Support Systems (DSS).**

Real-World Use Cases:

- **Financial analysis (e.g., tracking sales by product and region).**
- **Marketing insights (e.g., customer trends over time).**
- **Inventory management (e.g., stock levels across locations).**

OLAP vs. OLTP

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- Designed for analysis and reporting.
- Focuses on aggregated, summarized data.
- Examples: Monthly sales summaries, regional sales trends.

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- Designed for daily operations and data entry.
- Focuses on transactional, detailed data.
- Examples: Order processing, real-time customer transactions.

OLAP: Analytical, data summarization.

OLTP: Operational, data entry and updates.

Why Use OLAP Cubes?

Purpose of OLAP Cubes:

- Organize complex datasets into manageable, analyzable structures.
- Allow users to quickly view data from multiple dimensions (e.g., sales by product, region, and time).

Advantages of OLAP Cubes

Speed: Data is pre-aggregated, enabling faster responses to queries.

Insightful Analysis: Provides a summarized view of data for strategic decision-making.

Flexible Reporting: Easily change perspectives (e.g., monthly vs. yearly sales) with different dimensions.

Key Concepts in OLAP Cubes

- <u>**Cube**</u>**: A multi-dimensional structure for storing and analyzing data.**

  Think of it as a cube with multiple sides, each representing a dimension of the data.
- <u>**Dimensions**</u>**: Categories or attributes for data grouping.**

  Examples: Time, Location, Product.
- <u>**Facts**</u>**: Core, measurable data points (quantitative).**

  Examples: Sales Amount, Units Sold, Profit Margin.
- <u>**Measures**</u>**: Aggregated data values derived from facts.**

  Examples: Total Sales, Average Price, Max Inventory Level.

6.4.2 Dimension and Fact

What is a Dimension?

- Allows data to be grouped and viewed from different perspectives.
- Provides structure for analyzing facts across various angles (e.g., time, geography).

A dimension is a category or attribute by which data can be organized and analyzed in an OLAP cube.

Examples of Dimensions:

- Time: Year, Quarter, Month, Day
- Product: Category, Brand, Model
- Location: Country, Region, City, Store

Real-World Example: For a retail business, the Time dimension can show how sales vary by season, month, or day.

What is a Fact?

- Represents the core metric or event of interest, such as sales or revenue.
- Used in combination with dimensions to analyze performance.

A fact is a quantitative piece of data used for measurement and analysis in an OLAP cube.

Examples of Facts:

- Sales Amount: Total revenue from sales transactions.
- Quantity Sold: Number of items sold.
- Cost: Total expense associated with a product or service.

Real-World Example: In a retail OLAP cube, a fact could be Total Sales, representing the cumulative revenue generated across products, time periods, and locations.

Facts and Dimensions in Action

Combining Facts and Dimensions:

- **Facts provide the *numbers*; dimensions provide the *context*.**
- **Together, they allow multi-dimensional analysis, answering questions like, “How much was sold?” and “Where and when were the highest sales?”**

Example Scenario:

- **Question**: "What were the total sales for each product category in 2023?"
- **Fact**: Total Sales
- **Dimensions**: Product (e.g., Category), Time (e.g., Year 2023)

Discussion

Think of a scenario in your own industry or a familiar context.

Identify a Fact and two Dimensions you would analyze (e.g., total transactions for a banking service across branches and time periods).

Measures in OLAP Cubes

Measures are numerical values in an OLAP cube that quantify a specific fact, providing key metrics for analysis.

- **Measures are quantitative data points in an OLAP cube used for analysis.**
- **Represent the metrics that can be calculated or aggregated across dimensions.**
- **Examples: Total Sales, Average Order Value, Quantity Sold.**

Example of Measures in Action:

- **Measure: Total Sales**
- **Dimensions: Product Category, Time, Location**
- **Analysis Question: “What were the total sales for each product category by month and region?”**

Measures in OLAP Cubes

Continued...

Why Measures Matter:

- Enable data analysis and performance tracking.
- Allow multi-dimensional analysis, revealing trends and insights.

Types of Measures:

- Additive: Can be summed across any dimension (e.g., Total Sales).
- Semi-Additive: Can be summed across some dimensions but not others (e.g., Account Balance by product but not over time).
- Non-Additive: Cannot be summed (e.g., Ratios, Percentages).

Key Takeaway: Measures provide the core metrics that, when combined with dimensions, unlock deep insights into business performance.

How OLAP Cubes Use Dimensions and Facts

Data Structure in an OLAP Cube:

- **Dimensions: Define the structure of the cube and enable drill-downs.**
- **Facts/Measures: Stored in the center of the cube, holding the quantitative data (e.g., sales figures).**

Multi-Dimensional Analysis: OLAP cubes allow for data analysis across multiple dimensions simultaneously, providing insights from different perspectives.

Cube Example:

- **Dimensions**: Time, Product, Location
- **Fact**: Sales Amount
- **Analysis**: View sales across each product, in various locations, over different time periods.

Real-World Example: For an e-commerce company, an OLAP cube can show sales trends by month, product category, and region to aid in marketing strategies.

Steps to Create an OLAP Cube

1. **Define the Business Requirements:**
    - Identify the key questions that need answering (e.g., "What product has the highest sales in each region?")
    - Determine which dimensions and facts are relevant.
2. **Identify the Data Source:** Select the source databases, tables, or files containing the data.
3. **Define Dimensions and Hierarchies:** Decide on dimensions (e.g., Time, Product, Location) and their hierarchies (e.g., Year > Quarter > Month).
4. **Choose Facts/Measures:** Select quantitative metrics (e.g., Sales Amount, Profit, Units Sold) to include as facts.
5. **Build and Populate the Cube:** Use a BI tool (e.g., Tableau) to create the OLAP cube and load the data.
6. **Validate and Test the Cube:** Ensure that the cube accurately reflects the data and meets analytical requirements.

Dimension Hierarchies

- **Enhanced Analysis: Drill down from summary to details.**
- **User-Friendly Navigation: Simplifies complex data views.**

A hierarchy organizes dimension data from general to specific, allowing for detailed analysis. Provides a logical drill-down path (e.g., Year > Quarter > Month > Day).

Examples of Common Hierarchies:

- **Time**: Year > Quarter > Month > Day
- **Geography**: Country > State > City > Store
- **Product**: Category > Brand > Model

Discussion

Think of a hierarchy relevant to your work or industry and list the levels
(e.g., Organization > Department > Team).

Questions and Discussion

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Last modified: 09 Oct 2024